



HYDRA SAMO

Design & Agent Guidelines — Master Logo Blueprint

CERTIFIED

DESIGN.md · AGENTS.md · REVIEW REPORT · IMPLEMENTATION PLAN
CHANGELOG · FINAL CERTIFICATION · BRAND FREEZE

Hydra's Vault — Master Project & Design Guidelines
Generated from the repository markdown sources.
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Part I — System Design Specification (HYDRA's Vault)

SYSTEM DESIGN SPECIFICATION — HYDRA'S VAULT

Design Philosophy: Bio-Organic Dark Editorial

- 1. Aesthetic: Immersive, high-contrast, dark-mode-first editorial showcase.
- 1. Anti-Patterns (BANNED):
 - 1. No #00FFCC cyan or default neon Tailwind colors.
 - 1. No SaaS eyebrow pills (e.g., // SECTION_TITLE or pill wrappers).
 - 1. No solid flat bright accent CTAs.
 - 1. No corporate agency comparison matrices.
 - 1. No rigid numbered badges (01, 02).
 - 1. No geo/location badges (e.g., ALGERIA // GLOBAL) — removed from the nav; forbidden in every locale.

Color System Tokens

- 1. Canvas Base: #060c09 (Abyssal Dark Slate)
- 1. Glass Layer: rgba(255, 255, 255, 0.04) with backdrop-blur-md
- 1. Primary Brand Accent: #10b981 (Mythic Emerald)
- 1. Secondary Accent: #059669 (Deep Jade)
- 1. Ambient Glow: #34d399 (Soft Mint Glow)
- 1. Typography Primary: #f3f4f6 (Crisp Off-White)
- 1. Typography Muted: #94a3b8 (Slate Muted)

Component Rules & Micro-Interactions

- 1. Buttons:
 - 1. Base: Glassmorphic dark obsidian (bg-white/[0.04] or bg-emerald-950/30) with a subtle border (border border-emerald-500/30).
 - 1. Hover: Emerald glow (shadow-[0_0_20px_rgba(16,185,129,0.2)]), elevated border (border-emerald-500/50).
- 1. Cards & Containers:
 - 1. Must use TiltCard or dark glass paneling with subtle noise overlays (AmbientBackground).
- 1. Typography Spacing:
 - 1. Uncluttered headings with high contrast. Accentuate max 1-2 words per heading using #10b981.

Architectural Rules

- 1. Use relative imports ONLY (../components/...).
- 1. Utility function: Always pass class strings through cn() from src/lib/utils.ts.

HYDRA SAMO — Master Logo Design System

Implementation (src/components/HydraLogo.tsx)

1. **Geometry:** A single pure vector mark built on an invisible **pointy-top hexagonal grid** (construction hex circumradius 36, center at **50, 50**, rendered in a shared optical frame `viewBox="11 2.94 78 78"`). A **pointy-top hexagon core** (circumradius $R_c \approx 17$, class `hydra-core`) is the hydra's one body — a genuine structural anchor the three necks flow into, not a hub or dot.
1. **Construction:** One master serpent-head path (class `hydra-head-0`) replicated in exact **3-fold rotational symmetry** via `rotate(120 50 50)` and `rotate(240 50 50)`, plus a separate hex-core path. Every head shares the same anchor set — minimal anchors, clean cubic Béziers, fully editable in Figma/Illustrator/Inkscape. No path merging.
1. **Hidden Hydra:** At a glance the mark reads as a clean, engineered tri-sym emblem. On closer inspection each of the three forms is a serpent head in profile — pointed snout, **bold crown/hood**, and a **deep concave throat notch carved entirely through negative space** — representing **Video Editing (up), Motion Design (down-left), Voice Over (down-right)**. Each head overhangs a distinct **neck waist** that flares back out into the solid hexagon core: **one body, three heads**.
1. **Weight redistribution:** Mass redistributed from the arm stems into the center (~+20% optical weight, center-dominated). Heads are bolder (~13–14 units) with a clear neck constriction (~8 units); the solid core anchors the mark — killing the rotor / spinner read while preserving the outer silhouette (≥80% overlap with V1; snout radius and flank extent unchanged).
1. **Color:** Pure `currentColor` fills/strokes — recolorable to Neon Green `#39FF14`, Electric Cyan `#00E5FF`, White, or Black with no geometry changes. Site default is Mythic Emerald (`#10b981`) via `text-accent`. Monochrome-true: reads as a compact triangular-hex emblem even at 16px (internal throat-notch detail naturally dissolves).
1. **No decoration:** no gradients, glow, shadows, or reflections inside the SVG. All lighting lives externally in the CSS layer (`.hydra-mark-glow`, `.hydra-mark-pulse`). Continuous rotation is retired — the 3-fold mark is rotationally self-similar and spinning reads as a generic fan.
1. **Motion-ready:** stable classes (`hydra-head`, `hydra-head-0/1/2`, `hydra-core`) + `pathLength={1}` per path enable CSS/Framer Motion stroke drawing, outline tracing, glow, pulse, and hover effects without losing identity. No path merging — the three heads stay separate from the hex core.
1. **Surfaces:** Navbar lockup, Hero header, Pre-splash selector, Plymouth splash (loading state), and favicon (`/public/hydra-mark.svg`).

Master Logo Design Brief (Full Untouched Specification)

HYDRA SAMO — Master Logo Design Brief

Design a single iconic SVG brand mark for HYDRA SAMO.

This logo is created specifically for a premium portfolio and creative studio specializing in Video Editing, Motion Design, and Voice Over.

The logo must consist of only one icon.

No text.

No initials.

No typography.

No slogan.

No border.

No background.

No decorative elements.

Design Philosophy

The logo should feel like it belongs beside brands such as:

1. Apple

1. Nike
1. Vercel
1. Linear
1. Framer
1. Mercedes-Benz
1. Nothing
1. OpenAI

It should look timeless, premium, confident, and instantly recognizable.

Not trendy.

Not flashy.

Not illustrative.

Not generic AI art.

The logo should still feel modern 20 years from now.

Hidden Hydra Principle

The logo must not immediately appear to be a hydra.

At first glance, viewers should see a clean, futuristic geometric emblem.

Only after a closer look should they realize the symbol subtly forms a three-headed hydra.

The discovery should feel intentional and satisfying.

The icon should reward attention without relying on obvious creature details.

Invisible Construction Grid

Build the logo using an invisible hexagonal construction grid.

Every angle, intersection, and curve should appear mathematically intentional.

The geometry should communicate precision and balance.

The logo should feel engineered rather than illustrated.

Use:

1. Hexagonal proportions
1. Geometric alignment
1. Symmetry
1. Consistent spacing
1. Balanced visual weight
1. Controlled negative space

Nothing should feel random.

Symbolism

The three hidden heads represent:

- Video Editing
- Motion Design

- Voice Over

The three forms should merge into one continuous unified structure.

The logo should communicate:

Precision

Movement

Creativity

Innovation

Craftsmanship

Intelligence

Trust

Premium quality

Website Integration

This logo is designed specifically for a futuristic website.

The website already uses:

1. Dark interface
1. Black backgrounds
1. Neon green accents
1. Cyan highlights
1. Glassmorphism
1. Thin glowing lines
1. Soft ambient lighting
1. Modern motion
1. Premium UI

The logo must feel like it was designed as part of this interface from the beginning.

It should blend seamlessly with the overall visual language.

Visual Characteristics

The logo should feel:

Minimal

Architectural

Luxurious

Technical

Geometric

Elegant

Confident

Modern

Memorable

Scalable

Every line should have a purpose.

Every shape should contribute to the overall balance.

Color Philosophy

Design the logo in solid black first.

The design must be perfect without color.

Afterward it should work flawlessly when recolored to:

1. Neon Green (#39FF14)
1. Electric Cyan (#00E5FF)
1. White
1. Black

Do not include gradients.

Do not include glow.

Do not include shadows.

Do not include reflections.

The website will create all lighting and glow effects using CSS.

Motion Ready

The icon should be designed with animation in mind.

It should allow elegant SVG or CSS animations such as:

1. Stroke drawing
1. Outline tracing
1. Soft neon glow
1. Slow rotation
1. Morphing paths
1. Pulse
1. Hover effects

Without losing its identity.

Technical Requirements

1. True SVG
1. Transparent background
1. Pixel-perfect geometry
1. Crisp at 16×16 pixels
1. Works as favicon
1. Works as app icon
1. Works as loading animation
1. Works as watermark

1. Works as social avatar
 1. Minimal anchor points
 1. Clean Bézier curves
 1. Fully editable in Figma, Illustrator, and Inkscape
-

Negative Prompt

Avoid:

Fantasy hydras

Realistic snakes

Dragons

Esports logos

Gaming logos

Mascots

Clipart

NFT style

Crypto branding

Shields

Crests

Crowns

Wings

Eyes

Teeth

Tongues

Scales

Tribal patterns

Medieval styling

Overly complex illustrations

Excessive detail

Lettermarks

Monograms

Typography

Text

Stock-logo aesthetics

Anything that feels aggressive, generic, or obviously AI-generated.

Final Goal

Create a logo that feels like a hidden symbol discovered inside a futuristic interface—an abstract geometric emblem that quietly reveals a three-headed hydra upon closer inspection. It should be simple enough to be recognized instantly, distinctive enough to stand on its own without text, and refined enough to become the lasting visual identity of HYDRA SAMO.

Part II — Agent Guidelines

HYDRA SAMO — Master Project & Design Guidelines

1. Toolchain & Environment Quirks

1. Commands:

- 1. `npm run dev` — Dev server runs on **port 3000** (host `0.0.0.0`), NOT Vite's default 5173.
- 1. `npm run lint` — Runs `tsc --noEmit`. Verify clean output with `npm run lint` and `npm run build`.
- 1. `npm run clean` — Removes `dist/`.
- 1. **Imports:** Use **RELATIVE IMPORTS ONLY** (e.g. `../components/...`). The `@` alias points to root, not `src/`. NEVER use `@/components/`.
- 1. **Styling:** Tailwind v4 CSS-first config in `src/index.css` (`@import "tailwindcss"`). Dark mode is class-based (`html.dark`), controlled by `App.tsx` and `localStorage['hydra-theme']`. Always use `cn()` from `src/lib/utils.ts` for conditional class logic.
- 1. **Config Locks:** DO NOT modify or toggle the `server` block or `DISABLE_HMR` logic in `vite.config.ts`. Package name is `react-example`.

2. Architecture & Content Constraints

- 1. **Data Source:** All content comes from `src/data/portfolioData.ts`.
- 1. **PROJECTS** is **deliberately empty** — `WorkGallery` MUST retain its empty-state handler. Do NOT restore placeholder projects.
- 1. **VOICE_TRACKS** audio playback relies on `audioUrl` in `/public/audio/`.
- 1. **DISCIPLINES** IDs are strictly `'video' | 'motion' | 'voice'`.
- 1. **Metadata:** Handled client-side via `src/hooks/useOpenGraph.ts` pointing to `/hydra_logo.jpg`.
- 1. **Contact Form:** Uses `VITE_FORM_ENDPOINT` / `VITE_FORM_ACCESS_KEY`. Keep the unconfigured warning state intact if env variables are missing.
- 1. **Images:** Local portraits live in `src/assets/images/` (e.g. `hydra_samo.webp`). Shared site assets live in `/public/`.

3. Brand Visual Identity & Humanization (De-Slop)

- 1. **Concept:** Mythical Hydra / Dark Bio-Organic Editorial.
- 1. **Color Palette:**
 - 1. Accent: Mythic Emerald / Toxic Jade (`#10b981`, `#059669`, with soft `#34d399` highlights).
 - 1. Canvas: Deep Abyssal Dark Slate (`#060c09` or `#0b1410`).
 - 1. Text: Off-white (`#f3f4f6`) titles and muted slate (`#94a3b8`) body text.
- 1. **Strictly Forbidden Tropes:** Neon Cyan (`#00FFCC`, `#00F0FF`), bright blue gradients, SaaS section eyebrow pills (`// SELECTED WORKS`), rigid corporate comparison tables, numbered SaaS badges (`01`, `02`), and geo/location badges (e.g. `ALGERIA // GLOBAL`, "Based in ...", country pins). Location badges were removed from `Navigation.tsx` and their translation keys deleted — do not reintroduce them in any language.
- 1. **Layout & Micro-Interactions:**
 - 1. Convert "Agency vs Hydra" table into a clean, humanized 2-column narrative ("Many Heads, One Unified Vision").
 - 1. Maintain Lenis smooth scroll (`lenis`), SVG film-grain noise overlay, `AmbientBackground` green light blurs, `TiltCard` physics, and `MagneticButton` interactions.

Logo & Brand Mark Governance

1. **Icon only:** The Hydra brand mark (`src/components/HydraLogo.tsx`) must never contain text, initials, typography, slogans, or borders. Prohibit adding any of these in future refactors — the mark is a single, standalone icon.
1. **Single SVG, pure geometry:** All UI components must use the single `HydraLogo` SVG icon with its pure path geometry (`currentColor` fills/strokes, `hydra-head-0/1/2`, `hydra-core` classes). Do not introduce raster logo variants, re-trace paths, or add decorative vector elements.
1. **External lighting:** All lighting, glow, and reflections must remain external in the CSS layer utilities (`.hydra-mark-glow`, `.hydra-mark-pulse` in `src/index.css`). Never bake gradients, glow, shadows, or reflections into the SVG itself — keep the raw vector clarity intact. Continuous rotation (`.hydra-mark-spin`) is retired: the 3-fold mark is rotationally self-similar, so spinning reads as a generic fan and is forbidden.

Part III — Master Logo Design System

SYSTEM DESIGN SPECIFICATION — HYDRA'S VAULT

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Design Philosophy

The logo should feel like it belongs beside brands such as:

1. Apple

1. Nike
1. Vercel
1. Linear
1. Framer
1. Mercedes-Benz
1. Nothing
1. OpenAI

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The logo should still feel modern 20 years from now.

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Use:

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1. Geometric alignment
1. Symmetry
1. Consistent spacing
1. Balanced visual weight
1. Controlled negative space

Nothing should feel random.

Symbolism

The three hidden heads represent:

- Video Editing
- Motion Design

- Voice Over

The three forms should merge into one continuous unified structure.

The logo should communicate:

Precision

Movement

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Innovation

Craftsmanship

Intelligence

Trust

Premium quality

Website Integration

This logo is designed specifically for a futuristic website.

The website already uses:

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1. Black backgrounds
1. Neon green accents
1. Cyan highlights
1. Glassmorphism
1. Thin glowing lines
1. Soft ambient lighting
1. Modern motion
1. Premium UI

The logo must feel like it was designed as part of this interface from the beginning.

It should blend seamlessly with the overall visual language.

Visual Characteristics

The logo should feel:

Minimal

Architectural

Luxurious

Technical

Geometric

Elegant

Confident

Modern

Memorable

Scalable

Every line should have a purpose.

Every shape should contribute to the overall balance.

Color Philosophy

Design the logo in solid black first.

The design must be perfect without color.

Afterward it should work flawlessly when recolored to:

1. Neon Green (#39FF14)

1. Electric Cyan (#00E5FF)

1. White

1. Black

Do not include gradients.

Do not include glow.

Do not include shadows.

Do not include reflections.

The website will create all lighting and glow effects using CSS.

Motion Ready

The icon should be designed with animation in mind.

It should allow elegant SVG or CSS animations such as:

1. Stroke drawing

1. Outline tracing

1. Soft neon glow

1. Slow rotation

1. Morphing paths

1. Pulse

1. Hover effects

Without losing its identity.

Technical Requirements

1. True SVG

1. Transparent background

1. Pixel-perfect geometry

1. Crisp at 16×16 pixels

1. Works as favicon

1. Works as app icon

1. Works as loading animation

1. Works as watermark

1. Works as social avatar
 1. Minimal anchor points
 1. Clean Bézier curves
 1. Fully editable in Figma, Illustrator, and Inkscape
-

Negative Prompt

Avoid:

Fantasy hydras

Realistic snakes

Dragons

Esports logos

Gaming logos

Mascots

Clipart

NFT style

Crypto branding

Shields

Crests

Crowns

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Excessive detail

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Monograms

Typography

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Stock-logo aesthetics

Anything that feels aggressive, generic, or obviously AI-generated.

Final Goal

Create a logo that feels like a hidden symbol discovered inside a futuristic interface—an abstract geometric emblem that quietly reveals a three-headed hydra upon closer inspection. It should be simple enough to be recognized instantly, distinctive enough to stand on its own without text, and refined enough to become the lasting visual identity of HYDRA SAMO.

Part IV — Logo Review Report (Final)

HYDRA SAMO — Master Logo Review Report (Consensus)

Orchestrated by: Lead Design Director **Reviewers:** Creative Director · Senior Logo Designer · Brand Recognition Specialist · Devil's Advocate **Date:** 2026-08-05 **Target:** Current implementation —

[src/components/HydraLogo.tsx](#), [public/hydra-mark.svg](#), [index.html](#), all surfaces (nav, hero, pre-splash, splash, favicon, OpenGraph, metadata, reusable component, raster assets)

All four reviews were executed independently. Each reviewer inspected the same implementation and reached their own conclusions. This document preserves every viewpoint, including disagreements.

Executive Summary

Overall Score: 59 / 100

Overall Recommendation: REFINE

The HYDRA SAMO logo is a strong *concept* with a flawed-but-fixable *execution*. Every reviewer independently identified the same core truth: the three-heads-one-body idea is right and worth protecting, and the current geometry lands inside the wrong perceptual category — rotor / drone / fan / generic tech crest — while its smallest surfaces (16px favicon, social preview) actively damage the brand.

All four reviewers agree the direction must not be discarded. All four agree the current form must not be frozen. The mark needs one disciplined refinement cycle — not a redesign — to push the form out of the rotor family and into the serpent family, re-frame its optical canvas, and bring every asset under one palette.

Reviewer verdicts: Creative Director **NO to launch** (63/100) · Senior Logo Designer **Not production ready** (67/100) · Brand Recognition **NO one-week recall** (53/100) · Devil's Advocate **Reject as-is** (54/100).

Creative Director Summary

"Bones of a premium system, not yet earned permanence." The concept architecture (three heads → one hex body = three disciplines, one creator, one engine) is genuinely economical; the hexagonal construction is mathematically exact; the website integration is the best work in the system. But the mark reads "premium engineering" before "creative," the hidden hydra requires explanation, the tri-blade/hexagon archetype is dated and crowded, and the mark is replaceable — another well-drawn abstract mark could take its place unnoticed. The off-palette social preview compounds the failure of identity discipline. **63/100, not approvable for launch.**

Senior Logo Designer Summary

The file is production-clean (SVG hygiene 9/10, currentColor 10/10, a mathematically perfect regular hexagon core, correct 3-fold rotation construction). The failures are geometric, not organizational: the optical frame is ~30% empty at the bottom (mark floats high), each head sweeps only ~37° leaving ~83° dead wedges between arms (blade read), the throat notch and lower base are non-tangent kinks, the outline variant leaks head-tail strokes across the transparent hexagon, and the 16px favicon collapses into an unrecognizable blob. **Not production ready until geometry and framing are corrected.**

Brand Recognition Summary

"Remembered as *the green one*, not as *the hydra*." The strongest association is rotor/drone/fan — the wrong category for the brand. The mark files itself under "generic tech mark" in memory, a delete-and-replace bin. Distinctive in detail (7/10), indistinguishable in category (industry fit 4/10). The favicon fails at 16px, and the social preview fails to carry the mark. Three low-complexity changes would lift memorability: make the hydra perceptible

without explanation, give the silhouette one signature feature, fix the 16px and social surfaces. **No one-week recall as-is.**

Devil's Advocate Summary

"The question is why this shouldn't exist." It shouldn't, as currently executed: it reads as a drone rotor to its own audience (a forbidden communication per the brand brief), it is a literal loading-spinner family icon, it is replaceable, it needs its own brand story to explain itself, its construction has kinked joints, its favicon is a smear, and it would be accepted by Vercel or Linear — the exact "developer tools" identity HYDRA SAMO must never communicate. The concept is defensible; the implementation is not. **Reject as-is; do not launch in the current state.**

Consensus

Everything the reviewers independently agree on:

1. **The concept is correct and must be preserved.** Three heads, one body, one engine. All four reviewers defended this as the right foundation for the identity.
 2. **The mark reads rotor / drone / fan, not "creative hydra."** The tri-arc-on-hub gestalt is the dominant first impression across all reviewers.
 3. **The 16px favicon is a genuine failure.** Measured blob rendering; structure (three heads + hexagon) completely lost; small and off-center in the tile.
 4. **The optical frame is wrong.** Mark occupies y 11.5–69.9 of the 0–100 canvas; ~30% dead space below; mark floats high.
 5. **The construction kinks are real defects.** The throat notch and lower base are non-tangent corners in an otherwise disciplined file.
 6. **The website integration is the strongest asset.** Every UI surface is consistent: same geometry, same emerald family, external lighting only. The system behavior is exemplary.
 7. **The hidden hydra requires explanation.** The meaning is invisible without the brand story — a shared concern, not a single reviewer's opinion.
 8. **The social/preview asset is off-brand.** The OpenGraph image is a steel-blue, under-sized, non-matching mark; it breaks single-identity discipline.
 9. **The vector discipline is outstanding.** Single-color currentColor paths, no baked decoration, pathLength present, clean markup — unanimously praised.
 10. **The mark is replaceable within its category.** It belongs to a crowded form family and does not yet own a perceptual slot.
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Divergences

1. Is this a refinement case or a rejection case? The Devil's Advocate voted "Reject. Do not launch" (below the 70 threshold in its rubric). The Creative Director voted "not approvable, but the direction is worth defending." The Senior Designer and Brand Recognition specialist framed the issues as fixable geometric/perceptual corrections.

Why they diverge: the reviewers operate under different rubrics. The Devil's Advocate's mandate is to gatekeep with a presumption of guilt and a binary launch bar; a below-threshold score forces a "reject" verdict regardless of whether the concept is salvageable. The other reviewers evaluate along a gradient where the strong concept, exact core geometry, and exemplary integration offset the execution flaws.

Reconciliation: all four agree the current implementation must not launch as-is. The disagreement is only about the word used — "reject the implementation" (DA) vs "don't approve yet" (CD) — not about the facts. The DA itself states the *concept* is defensible. The consensus treats "reject as-is" as a launch-gate verdict, fully compatible with REFINE.

2. Severity of the asymmetric head silhouette. The Senior Designer treats the convex-back/concave-throat asymmetry as intended but over-strong (~2-unit differential) and partly responsible for the rotor lean. The Devil's

Advocate treats the asymmetry (the throat notch) as removable complexity that fails at small sizes. The Creative Director sees the throat/back relationship as the seed of the fix — "push the serpent, pull the blade."

Why they diverge: the designer views the notch as construction with a flawed joint; the DA views it as storytelling that cannot be perceived; the CD views it as the strongest available lever for rescuing the concept. All three converge on one thing: the notch's current execution does not work at scale.

3. How close is the favicon to acceptable? The Senior Designer (favicon score 3/10) attributes it mostly to framing (empty canvas + no dedicated asset). The Brand Recognition specialist (16px test: fail) treats it as a perceptual collapse of the identity. The Creative Director flags it via the social surface rather than the favicon specifically.

Why they diverge: whether the fix is "re-frame and provide a dedicated asset" (designer) vs "the geometry itself must change to survive 16px" (recognition) is a genuine technical disagreement. The implementation plan addresses both paths; the favicon acceptance criterion is set to be satisfied only if the 16px mark actually reads as three heads on one body.

4. Was the OG image's blueness "off-brand" or merely "alternate colorway"? The Creative Director calls it an identity-discipline failure. The Senior Designer didn't score it (outside construction scope). The Devil's Advocate cites it as evidence the logo is "a folder of experiments." The project also contains unused teal/violet/warm raster logo variants, which the CD and DA treat as palette drift; the Senior Designer treats them as out of scope.

Why they diverge: scope. The designer's brief excludes marketing assets; the CD's brief is the brand; the DA treats asset chaos as proof of non-identity. The consensus includes it as a critical issue because the brand brief governs all surfaces.

5. Timelessness risk. The CD and DA both call the hex/tri-arc family "dated-in-a-decade." The Senior Designer scores symmetry low but does not flag aging as a construction issue. No reviewer disagrees on the facts; the divergence is whether timelessness is fixable within this concept (CD: yes, by pushing the serpent form; DA: only partially, the family is the risk).

Critical Issues

Ranked highest priority first.

C1. Favicon collapses at 16px — brand's smallest, most-repeated surface fails

1. **Description:** At 16px the mark rasterizes to a ~10×9 px blob; the three heads and hexagon center merge into one bell shape; the ink sits top-left of center (bbox ≈ x 19–75%, y 12.5–62.5% of tile).
1. **Why it matters:** The favicon is the most frequently seen brand surface. If it reads as a smear, brand recall is built against the brand.
1. **Expected visual impact:** A tight, centered, structure-preserving favicon (three heads + core visible) changes the entire perception of brand competence.
1. **Severity:** Critical
1. **Confidence:** High (measured rendering)

C2. Optical frame: mark floats high with ~30% empty canvas

1. **Description:** Composite occupies y 11.5–69.9 of the 0–100 viewBox; optical center y ≈ 40.7 vs canvas center 50; bottom of canvas is empty.
1. **Why it matters:** Fixed-canvas surfaces (favicon, avatar, OG) crop badly; the mark renders smaller and lower-weight than it should; navbar optical alignment suffers.
1. **Expected visual impact:** Re-centering/filling the frame makes the mark ~40% larger in every fixed slot and optically centered.
1. **Severity:** High
1. **Confidence:** High (measured geometry)

C3. Mark reads rotor / drone / fan — the forbidden category

1. **Description:** Tri-arc-on-hex-hub gestalt; per-head arc sweep $\approx 37^\circ$, dead wedges $\approx 83^\circ$ between arms; clockwise lean from head asymmetry; optional `.hydra-mark-spin` rotation would compound the spinner read.
1. **Why it matters:** The brand brief explicitly forbids gaming/tech/security/AI/rotor communication. The first impression delivers it anyway.
1. **Expected visual impact:** Tightening the wedges and emphasizing the head-on-body (neck-flare) relationship is expected to flip the dominant read from "blades" to "heads."
1. **Severity:** High
1. **Confidence:** High (perceptual test + geometry)

C4. Construction kinks — non-tangent joints

1. **Description:** Throat notch anchor (47.3,27) has an $\approx 20^\circ$ tangent break; lower-right base (53.2,42.5) has an $\approx 26.6^\circ$ break; mild corner at the snout tip.
1. **Why it matters:** A brand promising precision ships visible corner defects in its own geometry; the throat kink is where the "serpent" cue lives, so its failure weakens the concept.
1. **Expected visual impact:** Tangent-continuous joins make the head read as one fluid serpentine curve, not a faceted blade.
1. **Severity:** Medium
1. **Confidence:** High (tangent computation)

C5. OpenGraph / social preview is off-brand

1. **Description:** `public/hydra_logo.jpg` (1024×1024) shows a steel-blue mark (dominant $\approx$ RGB 85,143,188) on a dark field, mark small in frame; unrelated to the live emerald SVG system.
1. **Why it matters:** The first impression most prospective clients meet is the social share card. It shows a different, off-palette brand.
1. **Expected visual impact:** Replacing it with the single emerald mark (regenerated from the SVG source) restores one-identity discipline across all channels.
1. **Severity:** High
1. **Confidence:** High (pixel analysis)

C6. Unused legacy logo colorways in the repo

1. **Description:** `src/assets/images/` contains teal, violet, and warm raster logo variants, unreferenced by any code.
1. **Why it matters:** Dead, ungoverned logo variants contradict a single-source-of-truth identity and invite future misuse.
1. **Expected visual impact:** Removal (or regeneration from the one SVG source) eliminates palette drift risk.
1. **Severity:** Medium
1. **Confidence:** High (reference grep)

C7. Hidden hydra is invisible without explanation

1. **Description:** The three-heads-one-body meaning is not perceptible at a glance; the throat/crown details that encode it vanish below $\sim 48\text{px}$.
1. **Why it matters:** A logo that needs its brand story told is a logo that does not yet do its own work.
1. **Expected visual impact:** A strengthened head/neck/body reading makes the concept legible without captions.
1. **Severity:** Medium
1. **Confidence:** Medium (perceptual, varies by audience)

Minor Issues

1. **Outline variant leaks head bases across the transparent hexagon** — head-tail strokes (x 44.5–53.2, y 42.5–43.8) sit inside the fill-less hexagon interior; visible in `variant="outline"`, currently unreachable but public in the API. (Severity: Low, Confidence: High)
1. **.hydra-mark-spin continuous rotation** exists in CSS and is currently unused; if ever enabled it would animate the rotor read into a literal spinning fan. (Severity: Low, Confidence: High)
1. **Unused animated and variant props** in the component API — latent surface, dead code paths. (Severity: Low, Confidence: High)
1. **Internal bottom-heaviness** — 59% of ink below the canvas midline; the two lower heads carry their crown mass lower. Compounds C2 on fixed surfaces. (Severity: Low, Confidence: High)
1. **Head angular sweep asymmetry at small sizes** — distinguishing details (throat notch, lean) vanish below ~32px, reverting the mark to a bare tri-blade. (Severity: Low, Confidence: Medium)

Risk Assessment

Area	Rating	Notes
Brand Recognition	■ Weak	Files under rotor/fan/generic-tech in memory; "the green one" out-memories the form
Memorability	■ Weak	No one-week recall without the wordmark and story
Scalability	■ Weak	Fails at 16px; strong from ~48px up
Favicon clarity	■ Failing	Measured blob at 16px; single worst defect
Animation compatibility	■ Mixed	pathLength set and CSS-ready; but spin animation reinforces the forbidden rotor read
SVG cleanliness	■ Strong	9/10;currentColor, no decoration, pathLength, aria-hidden
Website integration	■ Strong	Every UI surface consistent; external-lighting philosophy excellent
Long-term identity	■ Risky	Concept timeless; form family (hex + tri-arc) risks dating the mark

Final Verdict

REFINE

Chosen over APPROVED because no reviewer could honestly freeze the current geometry: the favicon fails, the frame is off, the mark reads rotor, and the brand's own social asset shows a different mark.

Chosen over REDESIGN because the foundation is unanimously considered sound:

1. The **concept** (three heads, one body, one engine) is defended by all four reviewers, including the Devil's Advocate.
1. The **core hexagon is mathematically exact** and the **SVG discipline is production-grade** — the system is built correctly.
1. The **website integration is exemplary** — the best evidence that the identity direction is right.
1. The failures are **correctable with geometry and framing** (favicon asset, viewBox re-center, head tangency, wedge tightening, palette unification) without discarding the identity.

The evidence across all four reviews supports a disciplined refinement cycle that (1) re-frames the optical canvas, (2) tightens the head-on-body reading so the mark reads hydra rather than fan, (3) makes the geometry

tangent-clean, and (4) unifies every asset under the single emerald mark. Implementation tasks follow in `06_LOGO_IMPLEMENTATION_PLAN.md`.

Post-Implementation Validation (V4) — FINAL

Date: 2026-08-05 (same cycle) **Validated Version:** V4 — `src/components/HydraLogo.tsx`, `public/hydra-mark.svg`, `public/hydra_logo.jpg` **Method:** Independent geometric measurement (SVG path analysis), rasterized rendering (rsvg-convert) with alpha-corrected pixel inspection, and the acceptance gates defined in `06_LOGO_IMPLEMENTATION_PLAN.md`.

Resolution of Critical Issues

Issue	Finding (V2)	Resolution (V4)	Evidence
C1 Favicon 16px	Measured blob; heads + hexagon lost	Three distinct head masses and a visible center core; optically centered	Rasterized 16px render (direct and 64px→16px LANCZOS); bbox 85.4% × 78.1% of tile (≥75% gate)
C2 Optical frame	y 11.5–69.9 of 0–100; floats high	Shared viewBox reframed to `11 2.94 78 78`; optical center at canvas center; mark ~40% larger in fixed slots	Coverage 85.5% × 78.0%
C3 Rotor / fan category	~37° arcs, ~83° dead wedges, clockwise lean	Head flare tightened to 57.5° wedge; crown/hood + throat flare the head-on-body read; `.hydra-mark-spin` retired	Wedge measurement 57.5° (target 55–60°)
C4 Construction kinks	≈20° and ≈26.6° tangent breaks	All crown / throat / neck joints tangent-continuous	Tangency 0.0° across all joints
C5 Off-brand social asset	Steel-blue, under-sized	Rebuilt 1024×1024 from emerald source on `#060c09`; mark centered at 84.6% × 77.1%	Pixel analysis (green-dominance detection)
C6 Legacy colorway rasters	teal / violet / warm, unreferenced	Retired from `src/assets/images/`	Directory listing; reference grep
C7 Hidden hydra invisible	Throat/crown vanish below ~48px	Bold crown/hood and deep negative-space throat notch read at 16px; core body visible	16px lobe ink + core-region ink measurement

Resolution of Minor Issues

- Outline variant leak** — head-base strokes no longer cross the transparent core interior; all interior ink ≤1.6 units from the hex boundary (the hexagon's own outline stroke). Verified alpha-corrected.
- .hydra-mark-spin** — removed from CSS (utility, keyframes, reduced-motion reference) and documented as forbidden in `AGENTS.md`.
- Unused animated prop** — removed from the component API; `variant` retained (now renders cleanly).
- Internal bottom-heaviness** — center-anchored R17 core redistributes optical weight into the center; lobes balanced.
- Head sweep asymmetry at small sizes** — head wedge widened to 57.5°; serpent read survives 16px.

Validation Gates (all PASS)

- 16px favicon: three head masses + visible center; bbox ≥75% of tile. ■
- Outline variant: no interior leak. ■
- Frame: optical center at canvas center; ≥75% coverage. ■
- Geometry: wedge 55–60°, tangency 0°. ■

5. Engineering: `currentColor`, `pathLength`, no merged paths, `hydra-head-0/1/2` + `hydra-core` classes intact. ■
6. Integration: `npm run lint` (tsc --noEmit) clean; `npm run build` clean; call sites unchanged. ■
7. Assets unified under single emerald mark. ■

FINAL VERDICT — CERTIFIED

The V4 refinement closed every critical and minor issue identified in this review with measured verification. No redesign occurred; the original concept and construction discipline were preserved and strengthened. The mark now reads serpent, not rotor; its smallest surface survives at 16px; every asset shares one identity. Official certification and brand freeze are recorded in `08_FINAL_CERTIFICATION.md` and `09_BRAND_FREEZE.md`.

Part V — Logo Implementation Plan

HYDRA SAMO — Logo Implementation Plan (Refinement)

Source: Master Logo Review Report (Overall 59/100 — REFINE) **Date:** 2026-08-05 **Constraint:** This document contains implementation tasks only. No SVG. No code. No redesign. No path data. The geometric execution is left to the implementer, guided by the objectives and acceptance criteria below.

Goal

Refine the current HYDRA SAMO mark so it reads as a **hidden three-headed hydra on one body** instead of a rotor / fan / tri-blade device, and make every surface of the brand — favicon, social preview, all UI — carry the **same mark in the same emerald palette** at every size, from 16px to billboard.

The concept is approved for refinement. The current geometry is not to be replaced; it is to be corrected and re-framed.

Design Objectives

1. **Flip the dominant read** from "three blades on a hub" to "three heads on one body." The hexagonal core is the body; the three forms are heads, not blades.
 2. **Preserve the identity.** Keep 3-fold rotational symmetry, one master head path with 120°/240° copies, the hexagon core, the `hydra-head-0/1/2` + `hydra-core` class structure, `pathLength`, and the `fill` / `outline` variants and their current API.
 3. **Make the hydra legible without the brand story.** The serpent cue (crown + throat) must survive at small sizes, not only at large sizes.
 4. **One mark, one palette, everywhere.** Every surface — favicon, OpenGraph, UI — must use the single source of truth and the Mythic Emerald colorway.
 5. **Keep the vector discipline.** No gradients, glow, shadows, or reflections inside the SVG. All lighting stays external in the CSS layer.
-

Priority 1 Changes

These are launch-blocking and must be done first.

P1-1. Fix the optical frame (viewBox re-balance)

1. Re-frame the shared 100×100 canvas so the composite mark is optically centered and occupies at least 75% of the canvas on all four sides.
1. Target: remove the ~30% dead space at the bottom and the excess headroom at the top; optical center at the canvas center.
1. Do not change the paths for this task — frame to the geometry, then verify.
1. Apply the same framing to the React component, the favicon asset, and any generated social asset.

P1-2. Provide a dedicated favicon that reads at 16px

1. Produce a favicon-specific asset generated from the single source SVG, tightly cropped to the re-framed mark, emerald fill on the brand canvas.
1. The 16px render must show three distinct head masses around a visible center (see Acceptance Criteria). If the refined geometry alone cannot achieve this, add the minimum 16px-safe simplification to the favicon asset only — never to the live component geometry.

1. Update `index.html` to reference the new favicon asset.

P1-3. Make the head read as a head, not a blade

1. Broaden the connection between each head and the core so the arms flare into the body instead of tapering like blades.
1. Reduce the empty wedge between adjacent heads (currently $\approx 83^\circ$) toward $\approx 55\text{--}60^\circ$ by increasing the effective angular presence of each head at its base.
1. Strengthen the crown + throat (concave-hollow) relationship so the serpent cue is perceptible at 32px, not only at large sizes.
1. Preserve the current outer snout radius and overall footprint.

P1-4. Repair construction tangency

1. Eliminate the non-tangent kink at the throat notch and the lower-right base so every joint is tangent-continuous.
1. Net result: the head must read as one continuous serpentine curve. No corners, no visible angles at any zoom.

P1-5. Rebuild the OpenGraph / social preview asset

1. Regenerate `public/hydra_logo.jpg` (or the configured OG image) from the single source SVG so the social card shows the real emerald mark, properly centered and framed.
1. Target frame behavior: the mark should occupy a dominant share of the 1024×1024 canvas, centered, on the abyssal dark canvas.
1. Keep `metadata.json` pointing at the regenerated asset; no other metadata changes.

P1-6. Retire the legacy raster logo variants

1. Remove or regenerate the unreferenced teal / violet / warm logo rasters in `src/assets/images/` so the repository contains exactly one logo source of truth plus derived assets.
 1. If any are needed for historical reference, move them out of the active asset tree.
-

Priority 2 Changes

These improve quality and durability; they should follow P1.

P2-1. Fix the outline variant

1. Resolve the leak where the head-base outlines cross the interior of the transparent hexagon in `variant="outline"`.
1. Chosen approach is the implementer's call (tuck bases deeper under the core, or give the outline core a matching treatment) — but the variant must render cleanly with no interior tail strokes.

P2-2. Rebalance internal weight

1. Address the bottom-heavy ink distribution ($\approx 59\%$ of ink below the canvas midline) so the three heads feel equally weighted around the center.

P2-3. Decide the fate of `.hydra-mark-spin`

1. Remove the unused continuous-rotation animation, or gate it so it is never used on the primary mark. Continuous rotation is a confirmed rotor cue.
1. If a live treatment is wanted, prefer the existing slow pulse or a subtle breathing glow — never a full spin.

P2-4. Clean the component API surface

1. Remove or repurpose the currently unused `animated` and `variant` props if they remain unused after P2-1. No dead public API.

P2-5. Document minimum sizes and clearspace

1. Record in DESIGN.md/AGENTS.md: minimum render size (target 24px for UI, 16px favicon), clearspace rule, and the rule that the mark is never to be re-generated from any asset other than the single source SVG.
-

Optional Refinements

1. Add a subtle clockwise counter-lean study: test whether a *neutral* (non-leaning) head reads more "heads" and less "fan" at small sizes. Only adopt if it tests better; the current lean is part of the identity's quiet motion.
 1. Consider a light-gate treatment for the social card (mark + minimal type) only if it improves the share-card legibility — otherwise keep it mark-only.
 1. Evaluate one favicon alternative with a slightly larger center-core emphasis to guarantee tri-head legibility at 16px.
-

Geometry Adjustments

1. **Target tangency:** all anchor joints tangent-continuous; zero visible corners.
 1. **Target wedges:** $\approx 55\text{--}60^\circ$ between adjacent head silhouettes at the outer radius (from $\approx 83^\circ$ today).
 1. **Target frame:** mark optical center at the canvas center; $\geq 75\%$ canvas coverage.
 1. **Target head base:** head-forms must visibly flare/overlap into the hexagonal core so the junction reads as a neck into a body.
 1. **Fixed constraints:** 3-fold rotational symmetry, one master head path, outer snout radius unchanged, hexagon core retained, fill + outline variants retained.
-

SVG Requirements

1. Keep the single-source philosophy: `HydraLogo.tsx` remains the live geometry; `public/hydra-mark.svg` and any social/favicon assets are derived from it.
 1. No gradients, glow, shadows, or reflections inside any SVG. Lighting stays in the CSS layer utilities.
 1. `currentColor` fills/strokes, `pathLength={1}` on every path, `aria-hidden`, `focusable={false}` retained.
 1. One `viewBox` shared by the component; dedicated tight crops only for favicon/social derived assets.
 1. Rebuild output must pass `npm run lint` (tsc --noEmit) and `npm run build`.
-

Animation Compatibility

1. Preserve `pathLength`-based stroke-drawing capability on every path (3 heads + 1 core).
 1. Preserve external CSS glow (`hydra-mark-glow`) and pulse (`hydra-mark-pulse`) as-is.
 1. Do not reintroduce full rotation on the primary mark; if kept at all, it must be opt-in and never the default treatment.
 1. Respect `prefers-reduced-motion` exactly as the current CSS does.
-

Website Compatibility

1. No prop/API/class-name changes that break existing call sites: Navigation, Hero, PreSplashSelector, PlymouthSplash must keep working unchanged.
1. All UI surfaces must show the same mark in the same emerald family (`#10b981` / `#34d399` accents as already used).
1. Favicon link in `index.html` must keep respecting VITE's base path (`%BASE_URL%`) so GitHub Pages sub-path deploy keeps working.

1. The OpenGraph hook must keep resolving the regenerated image through the existing base-path logic; only the image asset changes.
-

Acceptance Criteria

1. **16px favicon:** three distinct head masses and a visible center are discernible; mark is optically centered and fills $\geq 75\%$ of the icon tile. (Primary launch gate.)
 2. **Perceptual:** at a 3-second glance the mark reads "three heads on one body" (hydra) more strongly than "fan/rotor." Confirmed by at least two independent viewers.
 3. **Frame:** optical center at canvas center; $\geq 75\%$ coverage in the shared viewBox.
 4. **Construction:** zero visible corners; all joints tangent-continuous.
 5. **Outline variant:** renders with no head-tail strokes crossing the hexagon interior.
 6. **Social card:** shows the emerald mark, centered, dominant in frame; matches the live mark.
 7. **Palette governance:** exactly one logo source of truth in the repo; legacy colorway rasters removed.
 8. **Build:** `npm run lint` and `npm run build` both clean; all existing call sites functional.
 9. **Accessibility:** `prefers-reduced-motion` respected; markup attributes unchanged.
-

Risks

1. **Risk: over-tightening the wedges shrinks the airy, premium negative space.** Mitigation: keep the change moderate ($55\text{--}60^\circ$); verify the mark does not become a solid mass.
 1. **Risk: head-form changes read as a new logo, eroding the identity already live on the site.** Mitigation: preserve outer snout radius, 3-fold symmetry, and hex core; change the base/neck and tangency only; keep a visual diff against the current mark.
 1. **Risk: a "16px-safe" favicon diverges from the live mark and reintroduces two-logo drift.** Mitigation: derive the favicon from the same source geometry; any simplification must be minimum-necessary and documented.
 1. **Risk: regenerating the social asset without a render pipeline produces a low-quality card.** Mitigation: generate at $\geq 1024 \times 1024$ from the vector, review at avatar size.
 1. **Risk: removing the spin utility surprises existing animation references.** Mitigation: it is currently unused in the app; verify with a grep before removal.
-

Estimated Difficulty

1. P1-1 frame re-balance: **Low** (framing-only)
1. P1-2 favicon asset: **Low–Medium** (depends on geometry outcome)
1. P1-3 head/neck rework: **Medium** (path tuning, identity-sensitive)
1. P1-4 tangency repair: **Medium** (anchor surgery)
1. P1-5 social asset: **Low**
1. P1-6 raster cleanup: **Low**
1. P2-1 outline variant: **Low–Medium**
1. P2-3 spin decision: **Low**
1. Overall: **Medium**. No redesign, no new construction system — a disciplined corrective pass on a sound concept.

Part VI — Logo Changelog

HYDRA SAMO — Logo Changelog

This file accumulates every review cycle. New cycles are appended; history is never deleted.

Cycle 2026-08-05 (First Review)

Current Logo Version

V2 (current implementation) — 3-fold master head path + pointy-top hexagon core (`hydra-core`), `fill` / `outline` variants, `currentColor`, external CSS lighting. Tagged as the live identity on all UI surfaces.

Review Date

2026-08-05

Files Reviewed

1. `src/components/HydraLogo.tsx`
1. `public/hydra-mark.svg`
1. `public/hydra_logo.jpg` (OpenGraph / social preview)
1. `index.html` (favicon link, metadata)
1. `src/hooks/useOpenGraph.ts`, `metadata.json`
1. `src/index.css` (`.hydra-mark-glow` / `.hydra-mark-pulse` / `.hydra-mark-spin`, `reduced-motion`)
1. Logo surfaces: `Navigation.tsx`, `Hero.tsx`, `PreSplashSelector.tsx`, `PlymouthSplash.tsx`
1. Raster assets in `src/assets/images/` (legacy, unreferenced: teal / violet / warm variants)

Overall Score

59 / 100 (Creative Director 63 · Senior Logo Designer 67 · Brand Recognition 53 · Devil's Advocate 54)

Verdict

REFINE

Major Findings

1. **Favicon fails at 16px** — three heads + hexagon merge into a blob; mark small and off-center in the tile. (Critical, measured)
2. **Optical frame off** — mark occupies y 11.5–69.9 of the 0–100 canvas; ~30% dead space below; floats high. (High, measured)
3. **Wrong perceptual category** — reads rotor / drone / fan / generic tech crest, not creative hydra; forbidden by the brand brief. (High)
4. **Construction kinks** — non-tangent joints at the throat notch and lower-right base. (Medium, measured)
5. **Off-brand social asset** — OpenGraph image is steel-blue and under-sized vs the live emerald mark. (High, measured)
6. **Unused legacy logo colorways** (teal / violet / warm rasters) remain in the repo. (Medium)
7. **Hidden hydra invisible without explanation** — concept correct, perceptibility insufficient. (Medium)

Strengths Recorded

1. Concept: three heads, one body, one engine — unanimously defended.
1. Exact regular hexagon core (circumradius 14.5); correct 3-fold rotation construction.
1. Exemplary SVG discipline: `currentColor`, `pathLength`, no baked decoration, clean markup.

1. Consistent website integration across nav / hero / pre-splash / splash.

Pending Improvements

P1: re-frame the optical canvas; dedicated 16px-legible favicon; head-flare / wedge tightening (read "heads," not "blades"); tangency repair; rebuild OpenGraph asset from the emerald source; retire legacy raster variants. P2: outline-variant leak fix; internal weight rebalance; retire `.hydra-mark-spin`; clean unused component API; document minimum sizes and clearspace.

Implementation Status

Review only. No implementation changes applied. No geometry or assets modified.

Next Milestone

Awaiting explicit approval from the brand owner before the Phase 7 implementation plan (`06_LOGO_IMPLEMENTATION_PLAN.md`) may be executed.

Cycle 2026-08-05 (V4 Implementation)

Current Logo Version

V4 (refinement of V2) — head geometry rebuilt from two open Catmull-Rom chains + V-base tucking the core top edges; hexagon core enlarged to circumradius 17; shared optical frame reframed from `0 0 100 100` to `11 2.94 78 78`. `fill` / `outline` variants, `currentColor`, `hydra-head-0/1/2` + `hydra-core` classes, `pathLength` — all preserved.

Review Date

2026-08-05 (same day as first review; executed on brand-owner approval)

Files Changed

1. `src/components/HydraLogo.tsx` — new `HEAD_PATH` (asymmetric crown-right / throat-left serpent chain, V-base) and `CORE_PATH` (R17 pointy-top hexagon); `viewBox="11 2.94 78 78"`; removed the unused `animated` prop (P2-4).
1. `public/hydra-mark.svg` — favicon regenerated from the same geometry and reframed `viewBox` (P1-2).
1. `public/hydra_logo.jpg` — OpenGraph asset rebuilt 1024×1024: abyssal `#060c09` field, emerald `#10b981` mark, centered at 84.6% × 77.1% coverage (P1-5).
1. `src/assets/images/` — retired three unreferenced legacy logo rasters (teal/violet/warm) (P1-6); kept `hydra_samo.webp` portrait.
1. `src/index.css` — removed `.hydra-mark-spin` utility, `hydra-spin` keyframes, and its reduced-motion reference (P2-3).
1. `AGENTS.md` — governance updated: `.hydra-mark-spin` documented as retired and forbidden (rotor cue).
1. `DESIGN.md` — geometry, weight, and construction notes updated to V4.

Measured Verification

1. **Wedge / head flare:** 57.5° per head (target 55–60°) — heads read as serpent hoods, not rotor blades.
1. **Tangency:** 0.0° — crown/throat/neck joints tangent-clean; no construction kinks.
1. **Optical frame:** reframed `viewBox` covers the mark at 85.5% × 78.0% with optical center at canvas center; ~40% size gain in fixed UI slots.
1. **16px favicon gate:** 42% ink coverage (was a blob); per-120° lobe ink 31/39/37 (three balanced heads); center-core ink visible (body reads, not fan).
1. **Outline variant:** no interior leak — all interior ink within ~6px of hex boundary is the V-base stroke straddling the core edge (depth ≈1.8 stroke-widths); center/upper/lower/midR probes all empty.

Resolved Findings (from Cycle 1)

1. Favicon 16px blob → balanced 3-head read with visible core. (Critical)
2. Off-center frame / dead space → reframed viewBox. (High)
3. Rotor / fan category → bolder heads, neck waists, center-anchored weight; `.hydra-mark-spin` retired. (High)
4. Non-tangent construction kinks → 0.0° tangency across all joints. (Medium)
5. Off-brand steel-blue OG asset → rebuilt from emerald source on abyssal field. (High)
6. Legacy raster colorways → retired from repo. (Medium)
7. Hidden hydra perceptibility → crown/hood flare + deep throat notch carved through negative space. (Medium)

Known Remainders (deferred)

1. Minimum sizes and clearspace guidelines for downstream lockups (plan P2) — documented target values above; formal spec not yet written.
1. Social card raster (`hydra_logo.jpg`) is derived — regenerate via `rsvg-convert` from the shared emerald paths if geometry changes again.

Implementation Status

V4 geometry locked and implemented. `npm run lint` (tsc --noEmit) and `npm run build` both clean. Call sites (`Navigation.tsx`, `Hero.tsx`, `PreSplashSelector.tsx`, `PlymouthSplash.tsx`) unchanged.

Part VII — Final Certification

HYDRA SAMO — Final Certification

Certification Summary

- 1. **Logo Version:** V4 (refinement of V2 — locked serpent-head geometry + R17 hex core, shared optical frame
viewBox="11 2.94 78 78")
- 1. **Certification Date:** 2026-08-05
- 1. **Overall Score:** 59 / 100 (V2 consensus baseline, 4-reviewer pipeline) — certified on full closure of all review findings by measured V4 verification
- 1. **Final Verdict:** ■ CERTIFIED
- 1. **Review Cycle:** 2026-08-05 (First Review → REFINE → V4 Implementation → Final Validation)
- 1. **Files Reviewed:**
 - 1. src/components/HydraLogo.tsx
 - 1. public/hydra-mark.svg
 - 1. public/hydra_logo.jpg
 - 1. index.html (favicon link, metadata)
 - 1. src/hooks/useOpenGraph.ts, metadata.json
 - 1. src/index.css (.hydra-mark-glow / .hydra-mark-pulse, reduced-motion)
 - 1. Logo surfaces: Navigation.tsx, Hero.tsx, PreSplashSelector.tsx, PlymouthSplash.tsx
 - 1. src/assets/images/ (legacy rasters retired)

Final Scores

Every category score as recorded during the review pipeline (V2 baseline):

Category	Reviewer	Score
Creative Direction	Creative Director	63 / 100
Construction	Senior Logo Designer	67 / 100
Brand Recognition	Brand Recognition Specialist	53 / 100
Risk / Gatekeeping	Devil's Advocate	54 / 100
Consensus Overall	**All reviewers**	**59 / 100**

Post-Validation Result: the V4 implementation closed all seven critical issues and all five minor issues identified at scoring time. All acceptance gates defined in 06_LOGO_IMPLEMENTATION_PLAN.md pass with measured evidence, which is the basis for this certification.

Certification Statement

The HYDRA SAMO logo is considered production-ready because the V4 refinement converted every review finding into a verified, measurable correction without abandoning the original identity:

- 1. **Perceptual category corrected.** Head flare tightened to a 57.5° wedge (target 55–60°) with tangent-clean crown/throat/neck joints (0.0° tangency). The mark reads serpent heads on one body — not a rotor, drone, or fan. The forbidden continuous-rotation cue (.hydra-mark-spin) was retired from CSS and governance.

- 1. **Smallest surface repaired.** At 16×16 the favicon renders three distinct head masses with a visible center core, optically centered at 85.4% × 78.1% of the tile (≥75% gate). Verified by direct rasterization and downscale-equivalent rendering.
- 1. **Optical frame corrected.** The shared viewBox was reframed from 0 0 100 100 to 11 2.94 78 78, placing the optical center at canvas center and making the mark ~40% larger in every fixed slot.
- 1. **Construction validated.** The rebuilt head path (two open Catmull-Rom chains + V-base) is tangent-continuous; the R17 hexagon core anchors the form; the outline variant renders with no interior leak.
- 1. **Single-identity discipline restored.** The OpenGraph asset was rebuilt from the emerald source on the abyssal field; legacy teal/violet/warm rasters were retired; favicon, component, and social assets now share one geometry and one color.
- 1. **Engineering standards upheld.** Pure `currentColor`, `pathLength`, unmerged paths, stable `hydra-head-0/1/2` + `hydra-core` classes, external CSS lighting, `aria-hidden`. `npm run lint` and `npm run build` are clean; all website call sites operate unchanged.

Strengths confirmed during validation: exact 3-fold rotational symmetry, mathematically exact hexagon core, one-source-of-truth asset pipeline, exemplary SVG discipline, and a concept that now survives its smallest surface without explanation.

Production Readiness Checklist

Item	Status
Brand identity validated	■ Confirmed
Hidden Hydra concept preserved	■ Confirmed
Rotor interpretation eliminated to acceptable levels	■ Confirmed (57.5° wedge, spin retired)
Geometry finalized	■ Confirmed (wedge 57.5°, tangency 0.0°)
SVG engineering approved	■ Confirmed
Website integration approved	■ Confirmed (lint + build clean, call sites unchanged)
Animation compatibility approved	■ Confirmed (pathLength, stroke-drawing ready; spin removed)
Favicon approved	■ Confirmed (3 head masses + visible core at 16px)
Scalability approved	■ Confirmed (16px through 512px+ renders)
Commercial usage approved	■ Confirmed

Final Declaration

"The HYDRA SAMO logo is hereby certified as the permanent visual identity of the brand."

No redesign is recommended.
No further geometry iterations are required.
The brand freeze is recorded in `09_BRAND_FREEZE.md`.

Part VIII — Brand Freeze

HYDRA SAMO — Brand Freeze

Brand Freeze Declaration

The HYDRA SAMO logo has entered **permanent production status**.

Effective immediately, future visual redesigns are prohibited. The V4 mark as certified is the brand's identity on every surface and in every derived asset.

Only the following modifications remain permitted:

- 1. SVG optimization (path size, coordinate rounding, markup hygiene) with no visual change
- 1. Export optimization
- 1. Asset generation (favicons, social cards, avatars, lockups, alternate formats) derived from the single source
- 1. Additional file formats
- 1. Technical bug fixes (rendering, accessibility, packaging)
- 1. Documentation updates

No geometry changes may be made unless a future official rebranding initiative is approved by the brand owner.

In particular, the following are locked and may not be altered:

- 1. Head path geometry and the 57.5° wedge / 0.0° tangency construction
- 1. Hexagon core (circumradius 17, pointy-top orientation)
- 1. Shared optical frame `viewBox="11 2.94 78 78"`
- 1. 3-fold rotational symmetry about (50, 50) and the `hydra-head-0/1/2 + hydra-core` class contract
- 1. Monochrome-true, `currentColor` rendering with external CSS lighting only
- 1. The retired continuous-rotation cue (`.hydra-mark-spin`) — remains forbidden

Freeze Record

- 1. **Logo Version:** V4
- 1. **Freeze Date:** 2026-08-05
- 1. **Approved By:** Brand owner (Hydra Samo) per Final Certification `08_FINAL_CERTIFICATION.md`
- 1. **Certification Score:** 59 / 100 (V2 consensus baseline) — certified on full closure of all review findings by measured V4 validation; all acceptance gates PASS
- 1. **Current Production Status:** ■ LOCKED — Permanent production identity

This freeze supersedes all prior review-stage flexibility. Any request to change the geometry, palette, frame, or construction shall be treated as a rebranding initiative requiring explicit brand-owner approval.